Interface Specification

Control System, Operator Chair

Doc. no.: 10000017057

Information Type: SA21-DCMS INTERFACE SPECIFICATION

Revision 00| November 21, 2025

Revision History

Rev.	**Change reference** **(Change no.)**	**Section**	**Description of change**	**Date**
00	150148 – ODN1 DCMS Upgrade		First Issue	November 21, 2025

Table of Contents

1 Introduction

1.1 Purpose and scope

The purpose of this document is to describe the operator chair control system <-> equipment control system interfaces.

NOTICE

Interface template is owned and distributed by operator chair control system responsible.

Interface excel documents are stored and maintained in octoplant.

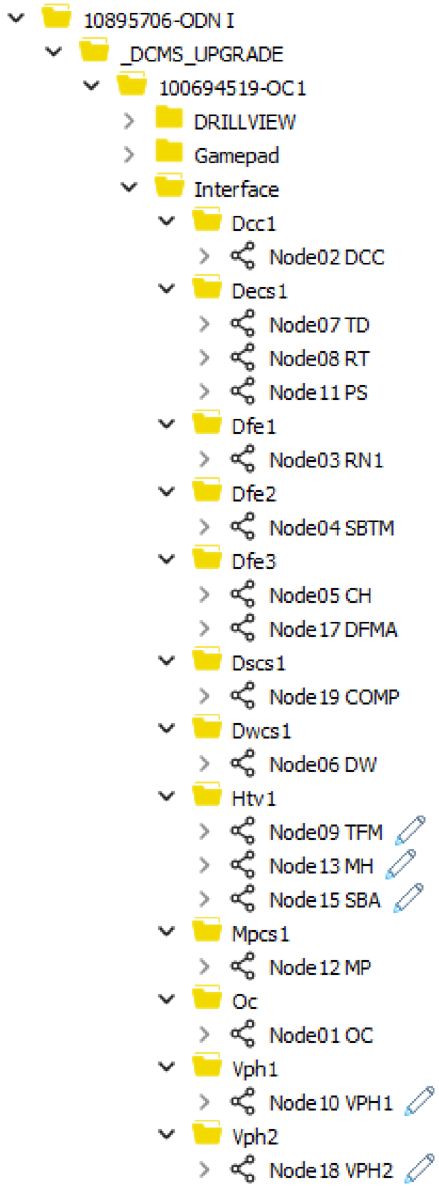


Figure 1 octoplant

2 Responsibility and Authority

2.1 Interface change request

Control system technical responsible.

2.2 Interface change approval

Operator chair technical responsible.

3 HMI Components

3.1 Function button

	HEX	DEC	Comments
Available			
NORMAL	B#16#0	0	DARK BLUE
CONFIRM	B#16#14	20	GREEN
NORMAL	B#16#1E	30	LIGHT BLUE
NORMAL	B#16#28	40	BLUE
	B#16#7F	127	INVISIBLE
Unavailable			
NORMAL	B#16#80	128	DARK BLUE
IBOP OPEN	B#16#94	148	GREEN
NORMAL	B#16#9E	158	LIGHT BLUE
NORMAL	B#16#A8	168	BLUE
FAULT	B#16#B2	178	RED
COM ERROR	B#16#B4	180	GREY

3.1.1 Invisible button status

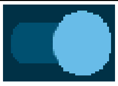
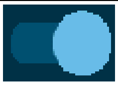
All buttons not in use (spare) shall send invisible status (DEC 127) to chair control system.

3.1.2 Additional button status

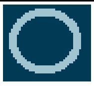
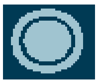
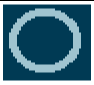
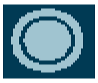
Any additional buttons status shall use even number status to show different text on the same button.

3.2 Settings

3.2.1 Toggle button

Button	Hex status	Dec status	Comments
	B#16#0	0	Available Not selected
	B#16#28	40	Available Selected
	B#16#80	128	Unavailable Not selected
	B#16#94	148	Confirm
	B#16#A8	168	Unavailable Selected

3.2.2 Radio button

Button	Hex status	Dec status	Comments
	B#16#0	0	Available Not selected
	B#16#28	40	Available Selected
	B#16#80	128	Unavailable Not selected
	B#16#94	148	Confirm
	B#16#A8	168	Unavailable Selected

3.2.3 Input field

	Comments
	Input available for operator
	Input unavailable for operator

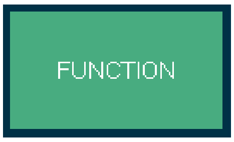
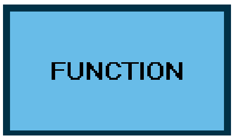
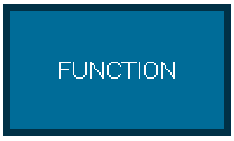
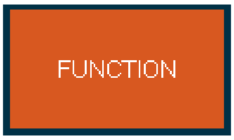
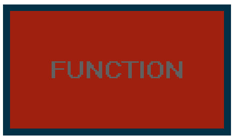
- Uses OcHmiParameterUdt

NOTICE

Possible to add confirm to activate/deactivate settings

4 Color Coding Philosophy

A common philosophy for colors and animations reflects the status of the equipment as follows:

Colour	Description
 Dark blue	Equipment is not running / deactivated / off, or not selected. Valve is closed. Buttons are available and ready for the operator's interaction. Pressing the button will trigger an action.
 Green	Function waiting for operator interaction. The selected function is awaiting and ready for confirmation. Press the CONFIRM button to start/activate the function.
 Light blue	Function is in progress (transition from one state to another). Start-up is in progress / valve is opening. Information. Pressing the button will normally not have any effect, but can in some cases be pressed to halt/stop running sequences/movements.
 Blue	Equipment is running / active / on, or selected. Valve is open. Buttons are available and ready for the operator's interaction. Pressing the button will trigger an action.
 Orange	Function requires operator's attention. Pressing the button will trigger an action.
 Red	Equipment or function has a failure warning / timeout. Start-up failure / valve not initialized. Information only and indicates an error condition. Pressing buttons will normally not have any effect.
 Grey	Communication error between operator chair and equipment. Signal error. Pressing buttons will normally not have any effect.
Invisible	Function not visible in HMI.

NOTICE

Grey text on the buttons indicates that the function is unavailable for the operator.

5 Mode Handling

5.1 Mode activation

1. Send Operator Id for active chair to all chairs.
2. Send activation status to mode button in selected chair.
3. Send unavailable status to mode buttons in all other chairs.

5.2 Mode deactivation

1. Send deactivation status to mode buttons in all chairs.
2. Set Operator Id to 0 in all chairs.

6 OFF-Signal

The OFF-signal is a signal sent from the operator chair control system if there is something wrong with the chair or if the operator presses Machines Off button in the touch-panel or Reset Chair button in DrillView.

OFF-signal is only sent to equipment selected in the actual chair.

OC Hardware Fault signal sent to equipment when there is a communication fault between OC PLC and HMI touch-panels or the joysticks.

6.1 OFF-signal without OC Hardware Fault signal

- Off-signal sent by the operator.
- Equipment:
  - Go to safe state.
  - Reset all chair modes.
  - Reset Operator Id; send Operator Id 0 to all chairs, making the equipment available in other chairs.

6.2 OFF-signal with OC Hardware Fault signal

- Off-signal sent by operator chair control system.
- Equipment:
  - Go to safe state, evaluate if the equipment should continue operation or not.
  - Reset all chair modes.
  - Reset Operator Id; send Operator Id 0 to all chairs, making the equipment available in other chairs.

7 Confirm Concept

To increase safety, the operator must press two buttons to execute a critical function, first the actual function button and then the Confirm button. This is referred to as the confirm process. All critical functions are operated this way to avoid unintended pushbutton operation and thereby accidents.

This functionality is implemented in two separate ways depending on the operation interface. Touch screen and joystick buttons are equipped with separate Confirm buttons.

7.1 Confirm sequence

- Operator presses a function with confirm, command is sent to equipment.
- Equipment sends back confirm status (148) to the function pressed and status 20 to CONFIRM button.
- Operator chair and equipment starts a 5 second confirm timer.
- Operator presses the confirm button and the command is sent to equipment.
- Equipment starts the function.
- Confirm not pressed within 5 second, equipment sends back the previous state to the function button, function not started/stopped.